EEG-RQVAE: Full Paper Title

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Abstract

1. Introduction

Electroencephalography (EEG) measures the brain's electrical activity through electrodes placed on the scalp, capturing oscillatory dynamics across a broad frequency spectrum with millisecond temporal resolution (Teplan, 2002; Müller-Putz, 2020). These properties make EEG a central modality in brain-computer interfaces (BCIs), with applications spanning motor imagery (Pfurtscheller & Neuper, 2001), imagined speech decoding (Elwasify et al., 2026), cognitive monitoring, and clinical diagnosis. Yet EEG signals are inherently noisy, non-stationary, and highly subject-specific, which makes decoding a notoriously difficult problem. Traditional approaches relied on hand-crafted features and shallow classifiers, but deep learning has substantially improved performance across tasks (Roy et al., 2019). Despite this progress, most methods remain task-specific and require large labeled datasets for each new application—a bottleneck that severely limits their clinical and real-world deployment.

The success of large pretrained models in natural language processing and computer vision has inspired a growing body of work on *EEG foundation models*—models trained on diverse, unlabeled EEG corpora and then adapted to downstream tasks via lightweight probing (Alain & Bengio, 2017). Several such models have been proposed recently, leveraging the Transformer architecture (Vaswani et al., 2017) with masked reconstruction as the pretraining objective: EEGPT (Wang et al., 2024), CBraMod (Wang et al., 2025a), REVE (Ouahidi et al., 2026), and CSBrain (Zhou et al., 2026). In parallel, state space models (SSMs) such as Mamba (Gu & Dao, 2024) and Mamba-2 (Dao & Gu, 2024) have emerged as efficient alternatives to attention-based architectures, with linear complexity in sequence length—a desirable property for long EEG recordings. EEGMamba (Wang et al., 2025b), FEMBA (Tegon et al., 2025), LUNA (Döner et al., 2025), and LuMamba (Broustail et al., 2026) explore this direction, but their pretraining objectives remain anchored to pixel-space reconstruction in the input signal domain. Concurrently, BAM-U-Net (Author, s) combines BiMamba-2 blocks with an adversarial denoising objective and achieves strong linear probing performance; however, it operates entirely in raw signal space and does not produce discrete, reusable token representations, nor does it incorporate any latent-space self-supervised objective.

We argue that reconstruction in the raw signal space is a suboptimal pretraining objective for learning *semantic* EEG representations. First, raw EEG contains substantial low-level noise and physiological artifacts that do not carry task-relevant information; a model trained to reconstruct them may allocate significant capacity to irrelevant details. Second, patch-based masked reconstruction encourages local pattern matching rather than higher-level semantic structure. VAE-based approaches for EEG (Zhao et al., 2024; You et al., 2025) alleviate some of these issues by learning a compressed latent space, but they have not been scaled to large corpora or combined with structured self-supervised pretraining. In other domains—audio compression with EnCodec (Défossez et al., 2022) and image tokenization with RQ-VAE (Lee et al., 2022)—residual vector quantization has demonstrated that discrete, hierarchical codebooks provide compact yet expressive representations that benefit downstream modeling.

In this work, we introduce **EEG-RQVAE**, a foundation model for EEG based on a Residual Quantized Variational Autoencoder trained with a two-stage pretraining pipeline designed to learn increasingly semantic representations. Inspired by MedVAE (Varma et al., 2025), which showed that decoupling reconstruction quality from semantic structure through

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a dedicated second training stage substantially improves the informativeness of the latent space in medical imaging, we adopt a similar principle for EEG. In the first stage, the model is trained to reconstruct a clean, augmented view of the EEG from a corrupted input, leveraging denoising as an implicit regularizer while a patch discriminator enforces signal fidelity. In the second stage, we adapt the encoder using a Joint-Embedding Predictive Architecture (Assran et al., 2023) in latent space, predicting the representations of masked spatial and temporal regions from visible context—encouraging the model to capture the structured, predictive dependencies that underlie brain dynamics. Throughout both stages, we use bidirectional Mamba blocks as the backbone, enabling efficient modeling of long EEG sequences while respecting the temporal symmetry of brain signals.

Our main contributions are:

- **EEG-RQVAE**: a residual vector-quantized VAE for multi-channel EEG that produces discrete, hierarchical token sequences suitable for both generative and discriminative downstream use.
- **A two-stage pretraining pipeline**: combining denoising reconstruction (Stage 1) with latent JEPA-style adaptation (Stage 2) to progressively enforce semantic representation learning.
- **An analysis of SSM architectures for EEG**: we investigate the relevance of bidirectional Mamba blocks for spatio-temporal EEG modeling and compare with Transformer baselines.
- **A fair evaluation benchmark**: we evaluate learned representations with a standardized linear probing protocol across multiple BCI tasks, with a particular focus on imagined speech decoding, and compare against recent EEG foundation models under identical conditions.

2. Related Work

EEG foundation models. Building universal EEG representations has been approached through two main architectural families. Transformer-based models (Vaswani et al., 2017) have dominated early work: EEGPT (Wang et al., 2024) combines masked reconstruction with a spatio-temporal representation alignment objective, while CBraMod (Wang et al., 2025a) introduces a criss-cross attention mechanism to jointly model spatial and temporal dependencies across channels. REVE (Ouahidi et al., 2026) scales masked pretraining to 25,000 subjects to learn topology-agnostic representations, and CSBrain (Zhou et al., 2026) adopts a cross-scale architecture to capture oscillations at multiple temporal resolutions. Despite strong performance under fine-tuning, Transformer-based models suffer from quadratic complexity in sequence length, which limits their applicability to long-duration clinical recordings. State space models have emerged as a linear-time alternative: EEGMamba (Wang et al., 2025b), FEMBA (Tegon et al., 2025), LUNA (Döner et al., 2025), and LuMamba (Broustail et al., 2026) all leverage Mamba-based backbones with masked reconstruction objectives. BAM-U-Net (Author, s) further integrates an adversarial patch discriminator into the denoising objective, yielding substantially stronger linear probing performance. However, all of these models share a common limitation: their pretraining objective operates in raw signal space, encouraging the model to faithfully reproduce waveform details that may be uninformative for downstream BCI tasks. Furthermore, none of them produce *discrete* token representations, which precludes their direct use in generative autoregressive pipelines.

Neural signal tokenization. Discrete representation learning has a long history in audio and image modeling. Residual Vector Quantization (RVQ) refines standard vector quantization by sequentially quantizing the residual error across multiple codebooks, achieving a favorable rate–distortion trade-off without increasing individual codebook size. EnCodec (Défossez et al., 2022) demonstrated that RVQ combined with adversarial training yields state-of-the-art audio compression, while RQ-VAE (Lee et al., 2022) showed that residual quantization of image features enables high-fidelity autoregressive image generation. In the EEG domain, LaBraM (Jiang et al., 2024) introduced a neural tokenizer based on vector-quantized spectral prediction, producing neural codes that serve as pretraining targets for a Transformer encoder. EEG-RQVAE extends this direction by combining RVQ with a two-stage pretraining pipeline applied directly to raw waveforms, without relying on pre-extracted spectral features.

Latent self-supervised learning and semantic representations. A growing body of work argues that self-supervised objectives formulated in latent space, rather than in signal space, yield more semantic representations. The Joint-Embedding Predictive Architecture (JEPA) (Assran et al., 2023) formalizes this intuition: instead of reconstructing masked patches from pixels, a predictor network learns to anticipate the latent representations of masked regions from visible context, preventing the model from expending capacity on low-level signal details. In the medical imaging domain, MedVAE (Varma et al., 2025) demonstrated that decoupling reconstruction quality from semantic structure through a dedicated second training stage—where the latent space is further refined using a vision-language model as a semantic anchor—significantly improves the utility of frozen representations for downstream classification. VAE-based approaches for EEG (Zhao et al., 2024; You et al., 2025) have confirmed that a compressed latent space captures clinically relevant features more compactly than raw signals, but these models have not been scaled to large multi-task corpora and their latent spaces are not shaped by any semantic self-supervised objective. EEG-RQVAE bridges these lines of work by (i) learning a discrete, hierarchically quantized latent space via RVQ, (ii) enforcing signal-level fidelity through adversarial denoising in Stage 1, and (iii) refining the encoder with a JEPA-style objective in Stage 2 that pushes representations toward semantic predictability without any reconstruction pressure.

3. Method

EEG-RQVAE is a foundation model for multi-channel EEG that learns discrete, hierarchically quantized representations through a two-stage pretraining pipeline. Figure 1 gives an overview of the full system. In Stage 1, an encoder–RVQ–decoder architecture is trained end-to-end to reconstruct a clean EEG window from a corrupted input, with an adversarial patch discriminator enforcing signal fidelity beyond what a pointwise loss can capture. In Stage 2, the decoder is frozen and the encoder is further adapted with a JEPA-style objective that operates entirely in discrete latent space, steering representations toward semantic predictability without any signal-level reconstruction pressure. Both stages share the same bidirectional Mamba-2 backbone. We first introduce the building blocks on which EEG-RQVAE relies, then describe the architecture and each training stage in detail.

3.1. Preliminaries

Selective state space models and bidirectional Mamba. State space models (SSMs) define a linear recurrence over a continuous input signal $u(t)$:

$$h'(t) = \mathbf{A}\,h(t) + \mathbf{B}\,u(t), \qquad y(t) = \mathbf{C}\,h(t), \tag{1}$$

where $h(t) \in \mathbb{R}^N$ is the latent state and $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are learned parameters. After zero-order-hold discretization with step size Δ, the recurrence becomes:

$$h_t = \bar{\mathbf{A}}\,h_{t-1} + \bar{\mathbf{B}}_t\,x_t, \qquad y_t = \mathbf{C}_t\,h_t, \tag{2}$$

with $\bar{\mathbf{A}} = e^{\Delta \mathbf{A}}$ and $\bar{\mathbf{B}}_t = \Delta\,\mathbf{B}_t$. Mamba (Gu & Dao, 2024) and its successor Mamba-2 (Dao & Gu, 2024) make $\mathbf{B}_t$, $\mathbf{C}_t$, and Δ input-dependent, yielding a *selective* SSM that can focus on or filter out individual time steps. Crucially, the recurrence in Eq. (2) is computed with a hardware-efficient parallel scan, achieving *linear* complexity $\mathcal{O}(T)$ in the sequence length T, in contrast to the $\mathcal{O}(T^2)$ cost of self-attention.

Because EEG windows are processed offline rather than as causal streams, future context is as informative as past context. We therefore adopt a bidirectional variant (Author, s) that processes the sequence in both directions and fuses the two outputs by summation. Denoting by $\mathcal{M}(\cdot)$ a single pre-normalized residual Mamba-2 block,

$$\mathcal{M}(X) = \mathrm{Mamba2}(\mathrm{LN}(X)) + X, \tag{3}$$

the BiMamba-2 block with sum fusion reads:

$$\mathrm{BiMamba}(X) = \mathcal{M}_{\mathrm{fwd}}(X) + \mathrm{rev}(\mathcal{M}_{\mathrm{bwd}}(\mathrm{rev}(X)))\,, \tag{4}$$

where $\mathrm{rev}(\cdot)$ reverses the sequence along the time axis. Each representation thus attends to both past and future temporal context while retaining linear-time inference.

Residual Vector Quantization. Vector Quantization (VQ) maps a continuous vector $z \in \mathbb{R}^d$ to its nearest entry in a finite codebook $\mathcal{E} = \{e_k\}_{k=1}^K \subset \mathbb{R}^d$:

$$q = \arg\min_k \|z - e_k\|_2. \tag{5}$$

Standard VQ is limited by the trade-off between codebook size K and representation precision. Residual Vector Quantization (RVQ) (Défossez et al., 2022; Lee et al., 2022) overcomes this by applying N quantization steps in cascade: the first codebook $\mathcal{E}^{(1)}$ quantizes the encoder output $z^{(0)} = z_e$, and each subsequent codebook $\mathcal{E}^{(n)}$ quantizes the residual left by the previous step:

$$q^{(n)} = \arg\min_k \left\| z^{(n-1)} - e_k^{(n)} \right\|_2, \qquad z^{(n)} = z^{(n-1)} - e_{q^{(n)}}^{(n)}, \tag{6}$$

yielding the quantized representation $z_q = \sum_{n=1}^{N} e_{q^{(n)}}^{(n)}$. Gradients are propagated through the quantization bottleneck via the straight-through estimator (Défossez et al., 2022), and the encoder is regularized by a commitment loss:

$$\mathcal{L}_{\text{commit}} = \| z_e - \text{sg}[z_q] \|_2^2, \tag{7}$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator. RVQ thus produces a sequence of N discrete indices per time step, forming a compact hierarchical token representation.

Joint-Embedding Predictive Architecture. JEPA (Assran et al., 2023) is a self-supervised learning framework that avoids signal-level reconstruction entirely. Given an input x, two views are created: a context x_c (visible patches) and a target x_t (masked patches). The context is encoded by an online encoder f_θ, and a lightweight predictor g_ϕ maps the context representation to a prediction of the target representation:

$$\hat{s}_t = g_\phi(f_\theta(x_c),\, m), \tag{8}$$

where m encodes the spatial position of the masked region. The target representation $s_t = f_{\bar{\theta}}(x_t)$ is produced by an exponential moving average (EMA) copy of the encoder with parameters $\bar{\theta} \leftarrow \tau\,\bar{\theta} + (1-\tau)\,\theta$, which provides a stable prediction target without requiring negative samples. The training objective minimizes the ℓ_2 distance between prediction and target in representation space:

$$\mathcal{L}_{\text{JEPA}} = \| \hat{s}_t - \text{sg}[s_t] \|_2^2. \tag{9}$$

Because the loss is computed entirely in latent space, the model is never penalized for ignoring high-frequency signal details irrelevant to the prediction task, which biases learning toward structured, semantic features.

3.2. EEG-RQVAE Architecture

Figure 1 gives an overview of EEG-RQVAE. A raw EEG window $x \in \mathbb{R}^{C \times T}$ ($C = 19$ channels, T time samples) is processed by four successive components: a channel adaptor, a spatio-temporal mixing block, a hierarchical convolutional encoder, and a residual vector quantizer. A symmetric decoder is used during Stage 1 only and is frozen for Stage 2.

Channel adaptor. EEG recordings vary in electrode count and montage across devices and datasets. To decouple the model's internal width from the raw channel count, we first pass the signal through a lightweight channel adaptor consisting of $L = 2$ point-wise convolutions (`Conv1d`, kernel size 1) interleaved with batch normalization and ReLU activations:

$$x' = \phi_{\text{adapt}}(x) \in \mathbb{R}^{C' \times T}, \quad C' = 64. \tag{10}$$

Because all convolutions have kernel size 1, the adaptor mixes channels but does not modify the temporal dimension.

Spatio-temporal mixing block. Before entering the hierarchical encoder, x' passes through a block that captures global spatio-temporal structure at the level of EEG patches. In the attention variant, this is a *CrissCross EEG Attention* module that operates on non-overlapping temporal patches of length $P = 200$ samples (one second at 200 Hz). Each patch is projected to a token of dimension d_{attn} and augmented with its short-time Fourier transform magnitude, providing a spectral view alongside the waveform. A positional encoding defined as a depthwise convolution over the (C, N) channel–patch grid is added before the tokens are split into two streams of equal width: a spatial stream that applies multi-head attention across the C channels at each patch position, and a temporal stream that applies multi-head attention across the N patches at each channel. The two streams are concatenated, processed by a feed-forward layer, and projected back to a per-sample residual that is added to x' through a learnable gate. In the Mamba variant, the CrissCross module is replaced by a single MambaEEGBlock applied to x', which attends to all time steps in a single pass.

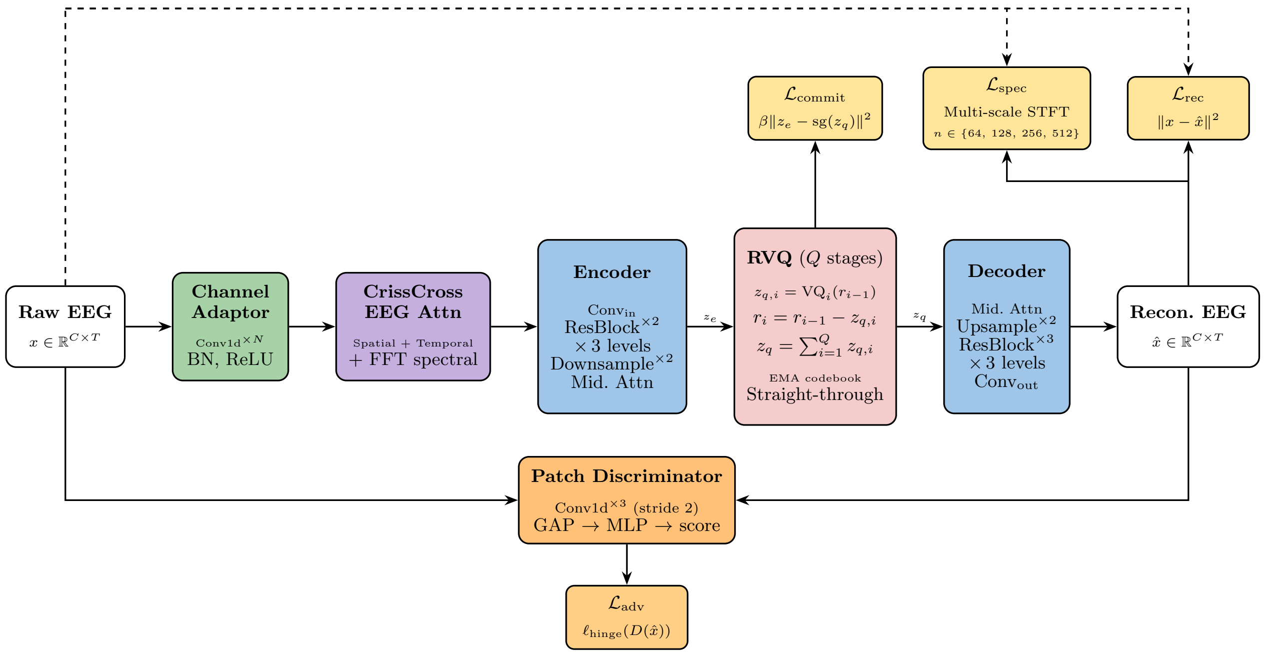

Figure 1. Overview of the EEG-RQVAE architecture. A raw EEG signal is first projected by a channel adaptor, then processed by a spatio-temporal mixing block (CrissCross attention or Mamba). A hierarchical encoder maps the sequence to a compact latent code z_e, which is discretized by the residual vector quantizer (RVQ, Q stages) into z_q. A symmetric decoder reconstructs the signal from z_q (Stage 1 only). Losses are colour-coded: reconstruction $\mathcal{L}_{\text{rec}}$, multi-scale spectral $\mathcal{L}_{\text{spec}}$, commitment $\mathcal{L}_{\text{commit}}$, and adversarial $\mathcal{L}_{\text{adv}}$.

Hierarchical encoder. The output of the mixing block is projected to $d_0 = 128$ feature maps by a `Conv1d` with kernel size 3, then processed by three resolution levels. At each level $\ell \in \{0, 1, 2\}$, two residual blocks expand the channel width:

$$d_\ell = 128 \times [1,\ 2,\ 4]_\ell = \{128,\ 256,\ 512\}, \tag{11}$$

and a strided convolution (stride 2) halves the temporal resolution at the first two levels. The signal is therefore downsampled by a factor of 4 along the time axis ($T \rightarrow T/4$). A bottleneck follows the last resolution level: two residual blocks sandwich a sequence block (Mamba or attention), providing global temporal context at the coarsest resolution. A final `Conv1d` projects to $z_e \in \mathbb{R}^{D \times (T/4)}$ with $D = 16$ via a `quant_conv` layer (kernel size 1).

Each residual block applies group normalization, a Swish activation, and two `Conv1d` layers (kernel size 3, padding 1) with a skip connection.

Residual vector quantizer. The encoder output z_e is passed to an RVQ with $N = 8$ stages, each stage using an EMA-updated codebook of $K = 512$ entries of dimension $D = 16$. The cascade produces the quantized representation $z_q = \sum_{n=1}^{N} e^{(n)}_{q^{(n)}} \in \mathbb{R}^{D \times (T/4)}$ and a set of N discrete index sequences $\mathbf{q} \in \{1, \ldots, K\}^{N \times (T/4)}$ that constitute the compact token representation of the EEG window. A `post_quant_conv` (kernel size 1) projects z_q before the decoder. The effective vocabulary over a single time step is $K^N = 512^8$, while each individual codebook lookup remains tractable. Codebook entries not used in a batch are replaced by a randomly sampled encoder output to prevent collapse (Laplace smoothing is applied to cluster sizes).

Decoder. The decoder mirrors the encoder: a bottleneck (residual blocks with sequence block) is followed by three resolution levels that progressively double the temporal resolution via nearest-neighbor upsampling and `Conv1d`, reaching the original length T. Each level uses num_res_blocks $+ 1 = 3$ residual blocks. A final `Conv1d` maps to $\hat{x} \in \mathbb{R}^{19 \times T}$. The decoder is trained jointly with the encoder in Stage 1 and frozen thereafter; only the encoder (and its channel adaptor) is retained for downstream probing.

3.3. Stage 1 — Denoising Reconstruction

The objective of Stage 1 is to train the full encoder–RVQ–decoder pipeline to produce physiologically plausible reconstructions of clean EEG from corrupted inputs, while building a compact discrete latent space. Figure 2 illustrates the two-stage pipeline.

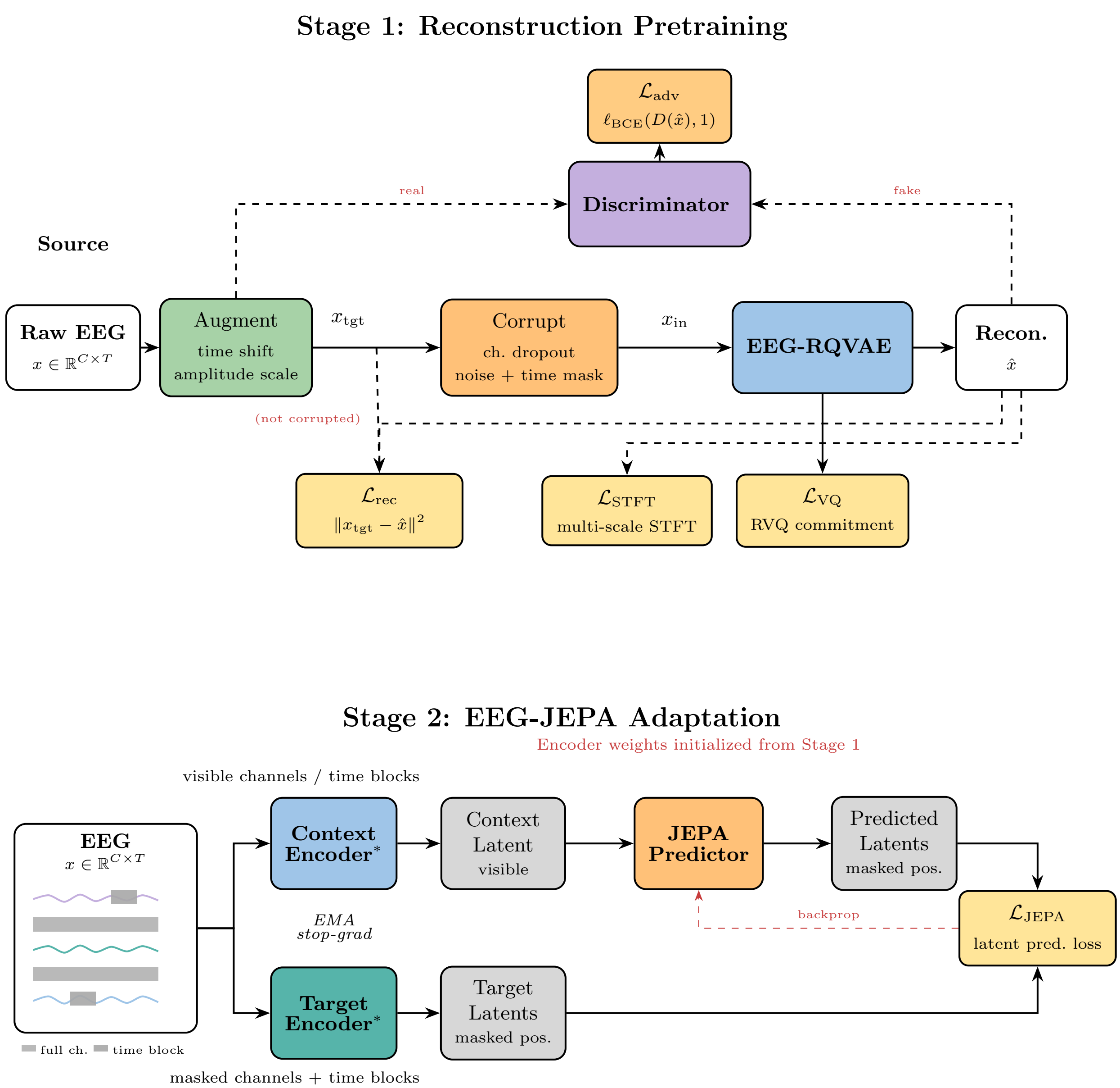

Figure 2. Two-stage pretraining pipeline. **Stage 1** (top): the raw EEG is first augmented to produce a clean target x_{tgt}, then independently corrupted to obtain the model input x_{in}. EEG-RQVAE reconstructs $\hat{x}$ from x_{in} and is trained to match x_{tgt}. **Stage 2** (bottom): the decoder is discarded; the encoder is adapted by a JEPA-style objective that predicts masked latent representations from visible context. * Encoder weights are initialized from Stage 1.

Data augmentation and corruption. Each training sample undergoes two independent stochastic transformations. First, a set of *augmentations* is applied to produce the *clean target* x_{tgt}: a random circular time shift (up to 10% of the window length), multiplicative amplitude scaling drawn from $\mathcal{U}(0.8, 1.2)$, dropout of up to 2 randomly selected channels (set to zero), additive Gaussian noise ($\sigma = 0.01$), and contiguous time masking of 5–10% of the sequence. Second, the augmented signal x_{tgt} is *corrupted* to produce the model input x_{in}: additional channel dropout and Gaussian noise are applied independently from the augmentation step, so that the model must recover meaningful structure rather than simply inverting a known transformation. This double-perturbation strategy encourages the encoder to learn representations that are invariant to the acquisition-level artefacts most commonly encountered in EEG, such as electrode impedance fluctuations and movement noise, while the decoder is trained to restore clean waveform morphology.

Reconstruction and spectral losses. The generator is trained to minimize a combination of a waveform loss and a multi-scale spectral loss between the reconstruction $\hat{x}$ and the *clean* target x_{tgt} (not the corrupted input x_{in}):

$$\mathcal{L}_{\text{rec}} = \text{SmoothL1}(\hat{x},\, x_{\text{tgt}}), \tag{12}$$

$$\mathcal{L}_{\text{spec}} = \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \mathcal{L}_{\text{L1}}\left(|\text{STFT}_n(\hat{x})|,\, |\text{STFT}_n(x_{\text{tgt}})|\right), \tag{13}$$

where $\mathcal{N} = \{64, 128, 256, 512\}$ are FFT window sizes (corresponding to 320 ms, 640 ms, 1.28 s, and 2.56 s at 200 Hz), covering all clinically relevant EEG frequency bands from delta to gamma. The multi-scale STFT loss penalizes spectral discrepancies at multiple temporal resolutions and is essential for preserving rhythmic EEG features that a pointwise loss tends to smooth out.

Adversarial loss. Starting from epoch 5, a patch-based discriminator D (NLayer-1D, $n_{\text{df}} = 64$, 3 strided convolutional layers) is introduced to enforce local temporal realism. The discriminator receives both the real signal x_{tgt} and the reconstruction $\hat{x}$ and is trained with the hinge loss:

$$\mathcal{L}_D = \frac{1}{2}\left(\mathbb{E}[\text{ReLU}(1 - D(x_{\text{tgt}}))] + \mathbb{E}[\text{ReLU}(1 + D(\hat{x}))]\right). \tag{14}$$

The generator is adversarially updated with:

$$\mathcal{L}_{\text{adv}} = -\mathbb{E}[D(\hat{x})]. \tag{15}$$

The adversarial objective forces the generator to produce high-frequency EEG patterns—such as sleep spindles or gamma-band oscillations—that are physiologically plausible but are typically smoothed out by the ℓ_1 or ℓ_2 reconstruction losses alone.

Total Stage 1 objective. The full generator loss combines the four terms:

$$\mathcal{L}_{\text{Stage}\,1} = \mathcal{L}_{\text{rec}} + \lambda_{\text{spec}}\,\mathcal{L}_{\text{spec}} + \lambda_{\text{commit}}\,\mathcal{L}_{\text{commit}} + \lambda_{\text{adv}}\,\mathcal{L}_{\text{adv}}, \tag{16}$$

with $\lambda_{\text{spec}} = 0.1$, $\lambda_{\text{commit}} = 1.0$, and $\lambda_{\text{adv}} = 0.01$. The commitment loss $\mathcal{L}_{\text{commit}}$ is defined in Eq. (7) and regularizes the encoder output toward the RVQ codebook.

3.4. Stage 2 — Latent JEPA Adaptation

Stage 2 refines the encoder representations toward semantic predictability. The decoder and the RVQ codebook are frozen; only the encoder (channel adaptor, convolutional layers, and bottleneck) is updated. No reconstruction objective is used. This clean separation mirrors the strategy of MedVAE (Varma et al., 2025), which showed that decoupling reconstruction quality from semantic structure through a dedicated second stage substantially improves downstream discriminability.

Model. The Stage 2 model consists of three components: (i) a *context encoder* (initialized from Stage 1) that processes the observed EEG window; (ii) a *target encoder*, an exponential moving average (EMA) copy of the context encoder whose parameters are never updated by gradient; (iii) a *narrow predictor* g_ϕ, a compact Transformer (4 blocks, $d_{\text{pred}} = 128$, 4 heads) that operates on the quantized latent sequence. The predictor first projects $z_q \in \mathbb{R}^{D \times T_z}$ from $D = 16$ to its internal dimension $d_{\text{pred}} = 128$, adds a learnable positional embedding, and applies 4 self-attention blocks before projecting back to $D = 16$. The "narrow predictor" design (Assran et al., 2023) is intentional: giving the predictor strictly less capacity than the encoder prevents it from trivially solving the prediction task without learning structure.

Latent masking. The temporal dimension of the quantized sequence $z_q \in \mathbb{R}^{D \times T_z}$ (where $T_z = T/4$) is divided into non-overlapping blocks of $B = 25$ tokens, corresponding to approximately 500 ms at 200 Hz with the encoder's $4\times$ temporal compression. At each training step, 25% of these blocks are randomly selected as the *target set* $\mathcal{T}$; the remaining blocks form the *context set* $\mathcal{C}$. Target positions in z_q are replaced by a shared learnable mask token $m \in \mathbb{R}^D$ (initialized from $\mathcal{N}(0, 0.02^2)$) before the context encoder, producing the masked sequence $\tilde{z}$. The target encoder always receives the complete, unmasked signal x.

JEPA objective. The predictor receives $\tilde{z}$ and the set of target positions $\mathcal{T}$ and produces predicted latent representations $\hat{s}_{\mathcal{T}} \in \mathbb{R}^{D \times |\mathcal{T}|}$. The target representations $s_{\mathcal{T}} = f_{\bar{\theta}}(x)_{\mathcal{T}}$ are extracted at the same positions by the EMA encoder. The Stage 2 loss is:

$$\mathcal{L}_{\text{Stage 2}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|\hat{s}_t - \text{sg}[s_t]\|_2^2 \,, \tag{17}$$

where $\text{sg}[\cdot]$ denotes stop-gradient. After each gradient step, the target encoder is updated by EMA with a cosine-scheduled momentum that warms up from $\tau = 1.0$ to $\tau = 0.999$, following the approach of Assran et al. (2023):

$$\bar{\theta} \;\leftarrow\; \tau\,\bar{\theta} + (1-\tau)\,\theta. \tag{18}$$

The high final momentum ($\tau = 0.999$) provides a stable prediction target and avoids the representational collapse that can occur when context and target encoders converge too rapidly. Because the loss in Eq. (17) is computed entirely in the discrete latent space z_q, the model is never penalized for ignoring the low-level signal details already handled in Stage 1, biasing learning toward the structured temporal dependencies that underlie EEG cognitive dynamics.

4. Experiments

4.1. Datasets and Preprocessing

Table 1. Datasets used for pretraining and downstream evaluation. Row 0 is used for pretraining only (no class labels). Rows 1–12 are evaluation datasets evaluated under linear probing.

BCI Task	Name	Ch.	Montage	Dur. (s)	Classes	Subjects	Samples
Pretraining (unlabeled)	TUH EEG Corpus	29	10-20 TCP (AR/LE)	30.0	—	1 385	303 686
Abnormal EEG detection	TUH Abnormal EEG	26	10-20 TCP (AR/LE)	30.0	2	1 164	68 486
Sleep staging	Sleep-EDF Expanded (50%)	2	Fpz-Cz / Pz-Oz	30.0	5	39	96 362
Emotion recognition	SEED-V	58	62ch 10-20	1.0	5	16	115 001
Motor imagery	PhysioNet EEG MMI	64	64ch 10-10	4.0	4	109	9 838
Emotion recognition	FACED	30	30ch 10-20	10.0	9	123	10 332
Motor imagery	BCI Competition IV-2a	22	22 EEG + 3 EOG	4.0	4	9	4 696
Seizure detection	Siena Scalp EEG	37	10-20	30.0	2	14	16 926
Seizure detection	CHB-MIT Scalp EEG	23	10-20 bipolar	10.0	1	24	353 872
Motor imagery	SHU-MI	32	32ch 10-20	4.0	2	25	11 988
Imagined speech (words)	BCI Competition 2020 T3	60	64ch 10-10	5.0	5	30	5 250
Imagined speech	KaraOne	58	64ch 10-20	5.0	11	14	1 913
Imagined speech	Chisco	62	122ch 10-10	4.0	39	5	32 403

4.2. Pretraining Setup

4.3. Linear Probing Evaluation

Table 2. Linear probing results across six downstream BCI tasks. Metrics reported: balanced accuracy (BACC), Cohen's κ (KAPPA), and macro-averaged F1 (MF1). Results are mean $\pm$ std over three random seeds. **Bold**: best. Underline: second best.

	BCI2A			PhysioNetMI			FACED			SEED-V			TUAB			Sleep-EDF		
Model	BACC	κ	MF1	BACC	κ	MF1	BACC	κ	MF1	BACC	κ	MF1	BACC	κ	MF1	BACC	κ	MF1
EEGPT (Wang et al., 2024)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CBraMod (Wang et al., 2025a)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
REVE (Ouahidi et al., 2026)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CSBrain (Zhou et al., 2026)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
LUNA (Döner et al., 2025)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
LaBraM (Jiang et al., 2024)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
BAM-U-Net (Author, s)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
EEG-RQVAE (ours)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

4.4. Ablation Study

5. Limitations

6. Future Work

7. Conclusion

References

Alain, G. and Bengio, Y. Understanding intermediate layers using linear classifier probes, 2017. URL https://openreview.net/forum?id=ryF7rTqgl.

Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y., and Ballas, N. Self-supervised learning from images with a joint-embedding predictive architecture, 2023. URL https://arxiv.org/abs/2301.08243.

Author(s), A. BAM-U-Net: A bidirectional adversarial mamba U-Net for large-scale EEG foundation modeling, 2026. URL https://anonymous.4open.science/r/BAM-U-Net/. Submitted to the 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Under review.

Balestriero, R. and LeCun, Y. Lejepa: Provable and scalable self-supervised learning without the heuristics, 11 2025.

Broustail, D., Tegon, A., Ingolfsson, T., Li, Y., and Benini, L. Lumamba: Latent unified mamba for electrode topology-invariant and efficient eeg modeling, 03 2026.

Coelli, S., Calcagno, A., Cassani, C. M., Temporiti, F., Reali, P., Gatti, R., Galli, M., and Bianchi, A. M. Selecting methods for a modular eeg pre-processing pipeline: An objective comparison. *Biomedical Signal Processing and Control*, 90:105830, 2024. ISSN 1746-8094. doi: https://doi.org/10.1016/j.bspc.2023.105830. URL https://www.sciencedirect.com/science/article/pii/S1746809423012636.

Dai, S., Xiao, H., Yu, S., and Ye, Q. Autoregressive visual decoding from eeg signals, 2026. URL https://arxiv.org/abs/2602.22555.

Dao, T. and Gu, A. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and Berkenkamp, F. (eds.), *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 10041–10071. PMLR, 21–27 Jul 2024. URL https://proceedings.mlr.press/v235/dao24a.html.

Döner, B., Ingolfsson, T. M., Benini, L., and Li, Y. LUNA: Efficient and topology-agnostic foundation model for EEG signal analysis. In *1st ICML Workshop on Foundation Models for Structured Data*, 2025. URL https://openreview.net/forum?id=TWaw5qtKQf.

Défossez, A., Copet, J., Synnaeve, G., and Adi, Y. High fidelity neural audio compression, 2022. URL https://arxiv.org/abs/2210.13438.

Elwasify, F., Shaaban, E., and Abdelmoneem, R. Eeg imagined speech neuro-signal preprocessing and deep learning classification. *Scientific Reports*, 16, 03 2026. doi: 10.1038/s41598-026-39395-6.

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., and Bengio, Y. Generative adversarial nets. In Ghahramani, Z., Welling, M., Cortes, C., Lawrence, N., and Weinberger, K. (eds.), *Advances in Neural Information Processing Systems*, volume 27. Curran Associates, Inc., 2014. URL https://proceedings.neurips.cc/paper_files/paper/2014/file/f033ed80deb0234979a61f95710dbe25-Paper.pdf.

Gu, A. and Dao, T. Mamba: Linear-time sequence modeling with selective state spaces. In *First Conference on Language Modeling*, 2024. URL https://openreview.net/forum?id=tEYskw1VY2.

Gu, A., Goel, K., and Ré, C. Efficiently modeling long sequences with structured state spaces, 10 2021.

Jiang, W., Zhao, L., and liang Lu, B. Large brain model for learning generic representations with tremendous EEG data in BCI. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=QzTpTRVtrP.

Lee, D., Kim, C., Kim, S., Cho, M., and Han, W.-S. Autoregressive Image Generation using Residual Quantization . In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11513–11522, Los Alamitos, CA, USA, June 2022. IEEE Computer Society. doi: 10.1109/CVPR52688.2022.01123. URL https://doi.ieeecomputersociety.org/10.1109/CVPR52688.2022.01123.

Ma, H., Chen, Y., Zhao, W., Yang, J., Ji, Y., Xu, X., Liu, X., Jing, H., Liu, S., and Yang, G. A mamba foundation model for time series forecasting. *CoRR*, abs/2411.02941, 2024. URL https://doi.org/10.48550/arXiv.2411.02941.

Müller-Putz, G. R. Chapter 18 - electroencephalography. In Ramsey, N. F. and del R. Millán, J. (eds.), *Brain-Computer Interfaces*, volume 168 of *Handbook of Clinical Neurology*, pp. 249–262. Elsevier, 2020. doi: https://doi.org/10.1016/B978-0-444-63934-9.00018-4. URL https://www.sciencedirect.com/science/article/pii/B9780444639349000184.

Ouahidi, Y. E., Lys, J., Thölke, P., Farrugia, N., Pasdeloup, B., Gripon, V., Jerbi, K., and Lioi, G. REVE: A foundation model for EEG - adapting to any setup with large-scale pretraining on 25,000 subjects. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026. URL https://openreview.net/forum?id=ZeFMtRBy4Z.

Pfurtscheller, G. and Neuper, C. Motor imagery and direct brain-computer communication. *Proceedings of the IEEE*, 89(7): 1123–1134, 2001. doi: 10.1109/5.939829.

Roy, Y., Banville, H., Carneiro de Albuquerque, I. M., Gramfort, A., Falk, T., and Faubert, J. Deep learning-based electroencephalography analysis: a systematic review. *Journal of Neural Engineering*, 16, 08 2019. doi: 10.1088/1741-2552/ab260c.

Samiee, S. and Baillet, S. Time-resolved phase-amplitude coupling in neural oscillations. *NeuroImage*, 159:270–279, 2017. ISSN 1053-8119. doi: https://doi.org/10.1016/j.neuroimage.2017.07.051. URL https://www.sciencedirect.com/science/article/pii/S1053811917306195.

Tegon, A., Ingolfsson, T. M., Wang, X., Benini, L., and Li, Y. Femba: Efficient and scalable eeg analysis with a bidirectional mamba foundation model. *CoRR*, abs/2502.06438, February 2025. URL https://doi.org/10.48550/arXiv.2502.06438.

Teplan, M. Fundamental of eeg measurement. *MEASUREMENT SCIENCE REVIEW*, 2, 01 2002.

Varma, M., Kumar, A., van der Sluijs, R., Ostmeier, S., Blankemeier, L., Chambon, P., Bluethgen, C., Prince, J., Langlotz, C., and Chaudhari, A. Medvae: Efficient automated interpretation of medical images with large-scale generalizable autoencoders, 2025. URL https://arxiv.org/abs/2502.14753.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L. u., and Polosukhin, I. Attention is all you need. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., and Garnett, R. (eds.), *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.

Wang, G., Liu, W., He, Y., Xu, C., Ma, L., and Li, H. Eegpt: Pretrained transformer for universal and reliable representation of eeg signals. In Globerson, A., Mackey, L., Belgrave, D., Fan, A., Paquet, U., Tomczak, J., and Zhang, C. (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 39249–39280. Curran Associates, Inc., 2024. doi: 10.52202/079017-1239. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/4540d267eeec4e5dbd9dae9448f0b739-Paper-Conference.pdf.

Wang, J., Zhao, S., Luo, Z., Zhou, Y., Jiang, H., Li, S., Li, T., and Pan, G. CBramod: A criss-cross brain foundation model for EEG decoding. In *The Thirteenth International Conference on Learning Representations*, 2025a. URL https://openreview.net/forum?id=NPNUHgHF2w.

Wang, J., Zhao, S., Luo, Z., Zhou, Y., Li, S., and Pan, G. Eegmamba: An eeg foundation model with mamba. *Neural Networks*, 192:107816, 2025b. ISSN 0893-6080. doi: https://doi.org/10.1016/j.neunet.2025.107816. URL https://www.sciencedirect.com/science/article/pii/S0893608025006963.

You, Z., Guo, Y., Zhang, X., and Zhao, Y. Virtual electroencephalogram acquisition: A review on electroencephalogram generative methods. *Sensors*, 25(10), 2025. ISSN 1424-8220. doi: 10.3390/s25103178. URL https://www.mdpi.com/1424-8220/25/10/3178.

Zhao, T., Cui, Y., Ji, T., Luo, J., Li, W., Jiang, J., Gao, Z., Hu, W., Yan, Y., Jiang, Y., and Hong, B. Vaeeg: Variational auto-encoder for extracting eeg representation. *NeuroImage*, 304:120946, 2024. ISSN 1053-8119. doi: https://doi.org/10.1016/j.neuroimage.2024.120946. URL https://www.sciencedirect.com/science/article/pii/S1053811924004439.

Zhou, Y., Wu, J., Ren, Z., Yao, Z., Lu, W., Peng, K., Zheng, Q., Song, C., Ouyang, W., and Gou, C. CSBrain: A cross-scale spatiotemporal brain foundation model for EEG decoding. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026. URL https://openreview.net/forum?id=agcXjEHmyW.

A. Foundation Models Used for Comparison

B. Ablation: Inductive Bias on Spatial Positioning of Channels

C. Computational Analysis

D. Implementation Details